LAND REFORMS, GREEN REVOLUTION AND CROPPING SYSTEMS

LAND-REFORM INSTRUMENT MAP → LAW-TO-OUTCOME RAIL → STATE REFORM COMPARISON → GREEN-REVOLUTION PACKAGE → CONCENTRATION LEGACY → CROPPING METRICS FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00 → 01 → 02 → 03 → 04 → 05 → 06 → 07 → 08 → 09 → 10 → 11 → E

STAGE 00 — LAND-REFORM INSTRUMENT MAP

LAND-REFORM INSTRUMENT MAP · INTERMEDIARY ABOLITION · CULTIVATOR-STATE LINK · RENT + SECURITY + RECORDED CULTIVATOR · PERMISSIBLE HOLDING + SURPLUS LAND · WORKABLE PARCELS + ENFORCEABLE RIGHTS · MUST REMEMBER · LAND REFORM, TENANCY, CEILINGS, CONSOLIDATION, IRRIGATION, SEEDS

INTERMEDIARY ABOLITION → cultivator-state link

TENANCY → rent + security + recorded cultivator → CEILINGS → permissible holding + surplus land → CONSOLIDATION / RECORDS → workable parcels + enforceable rights

CORE CLAIM
- INTERMEDIARY ABOLITION → cultivator-state link
- TENANCY → rent + security + recorded cultivator

MECHANISM
- CEILINGS → permissible holding + surplus land
- CONSOLIDATION / RECORDS → workable parcels + enforceable rights

CONSEQUENCE / LIMIT
- MUST REMEMBER: Land reform, tenancy, ceilings, consolidation, irrigation, seeds,...
- INTERMEDIARY ABOLITION → cultivator-state link

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Land reform, tenancy, ceilings, consolidation, irrigation, seeds,.

STAGE 01 — LAW-TO-OUTCOME RAIL

LAW-TO-OUTCOME RAIL · STATE LAW · ACCURATE RECORDS · LOCAL ADMINISTRATION + MOBILISATION · ACTUAL TENURE · REDISTRIBUTION OUTCOME

STATE LAW → ACCURATE RECORDS → LOCAL ADMINISTRATION + MOBILISATION → ACTUAL TENURE / REDISTRIBUTION OUTCOME

CORE CLAIM
- STATE LAW
- ACCURATE RECORDS

MECHANISM
- LOCAL ADMINISTRATION + MOBILISATION
- ACTUAL TENURE / REDISTRIBUTION OUTCOME

CONSEQUENCE / LIMIT
- STATE LAW
- ACCURATE RECORDS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LOCAL ADMINISTRATION + MOBILISATION; ACTUAL TENURE / REDISTRIBUTION OUTCOME.

STAGE 02 — STATE REFORM COMPARISON

STATE REFORM COMPARISON · OPERATION BARGA · SHARECROPPER RECORDING · TENANCY ABOLITION + CEILING DEPTH · J&K 1950 · EARLY ESTATE RESTRUCTURING · COMPARE CONTEXT; DO NOT COPY OUTCOMES MECHANICALLY

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
OPERATION BARGA → sharecropper recording	J&K 1950 → early estate restructuring	OPERATION BARGA → sharecropper recording
KERALA → tenancy abolition + ceiling depth	COMPARE context; do not copy outcomes mechanically	KERALA → tenancy abolition + ceiling depth

CORE CLAIM
- OPERATION BARGA → sharecropper recording
- KERALA → tenancy abolition + ceiling depth

MECHANISM
- J&K 1950 → early estate restructuring
- COMPARE context; do not copy outcomes mechanically

CONSEQUENCE / LIMIT
- OPERATION BARGA → sharecropper recording
- KERALA → tenancy abolition + ceiling depth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: J&K 1950 → early estate restructuring; COMPARE context; do not copy outcomes mechanically.

STAGE 03 — GREEN-REVOLUTION PACKAGE

GREEN-REVOLUTION PACKAGE · HYV SEED + ASSURED WATER · FERTILISER + CREDIT + EXTENSION · PROCUREMENT + PRICE SUPPORT · PACKAGE EFFECT · YIELD + MARKETED SURPLUS

HYV SEED + ASSURED WATER → FERTILISER + CREDIT + EXTENSION → PROCUREMENT + PRICE SUPPORT → PACKAGE EFFECT → yield + marketed surplus

CORE CLAIM
- HYV SEED + ASSURED WATER
- FERTILISER + CREDIT + EXTENSION

MECHANISM
- PROCUREMENT + PRICE SUPPORT
- PACKAGE EFFECT → yield + marketed surplus

CONSEQUENCE / LIMIT
- HYV SEED + ASSURED WATER
- FERTILISER + CREDIT + EXTENSION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROCUREMENT + PRICE SUPPORT; PACKAGE EFFECT → yield + marketed surplus.

STAGE 04 — CONCENTRATION LEGACY

CONCENTRATION LEGACY · IRRIGATED NORTHWEST · RICE SPECIALISATION · FOODGRAIN SURPLUS · GROUNDWATER + SOIL + REGIONAL STRESS

IRRIGATED NORTHWEST → WHEAT / RICE SPECIALISATION → FOODGRAIN SURPLUS → GROUNDWATER + SOIL + REGIONAL STRESS

CORE CLAIM
- IRRIGATED NORTHWEST
- WHEAT / RICE SPECIALISATION

MECHANISM
- FOODGRAIN SURPLUS
- GROUNDWATER + SOIL + REGIONAL STRESS

CONSEQUENCE / LIMIT
- IRRIGATED NORTHWEST
- WHEAT / RICE SPECIALISATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOODGRAIN SURPLUS; GROUNDWATER + SOIL + REGIONAL STRESS.

STAGE 05 — CROPPING METRICS FORK

CROPPING METRICS FORK · CROP-AREA DISTRIBUTION · GROSS CROPPED · NET SOWN AREA · ALLIED SHIFT · EACH NEEDS STATED TIME + GEOGRAPHY · CLOSE DISTINCTION · LAND RECORD IS NOT TITLE GUARANTEE, ABOLITION IS NOT

PATTERN → crop-area distribution

INTENSITY → gross cropped / net sown area → DIVERSIFICATION → crop / allied shift → EACH needs stated time + geography

CORE CLAIM
- PATTERN → crop-area distribution
- INTENSITY → gross cropped / net sown area

MECHANISM
- DIVERSIFICATION → crop / allied shift
- EACH needs stated time + geography

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Land record is not title guarantee, abolition is not complete...
- PATTERN → crop-area distribution

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Land record is not title guarantee, abolition is not complete.

STAGE 06 — DIVERSIFICATION DECISION BOARD

DIVERSIFICATION DECISION BOARD · AGRO-CLIMATE + WATER · EXPECTED RETURN + VOLATILITY · CREDIT + INSURANCE + EXTENSION · STORAGE + PROCESSING + BUYER ACCESS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
AGRO-CLIMATE + WATER	CREDIT + INSURANCE + EXTENSION	AGRO-CLIMATE + WATER
EXPECTED RETURN + VOLATILITY	STORAGE + PROCESSING + BUYER ACCESS	EXPECTED RETURN + VOLATILITY

CORE CLAIM
- AGRO-CLIMATE + WATER
- EXPECTED RETURN + VOLATILITY

MECHANISM
- CREDIT + INSURANCE + EXTENSION
- STORAGE + PROCESSING + BUYER ACCESS

CONSEQUENCE / LIMIT
- AGRO-CLIMATE + WATER
- EXPECTED RETURN + VOLATILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREDIT + INSURANCE + EXTENSION; STORAGE + PROCESSING + BUYER ACCESS.

STAGE 07 — MILLET TRANSITION RAIL

MILLET TRANSITION RAIL · DRYLAND SUITABILITY · LOWER WATER EXPOSURE · RESILIENCE VALUE · NEED PROCESSING + PROCUREMENT + DEMAND

DRYLAND SUITABILITY → LOWER WATER EXPOSURE → NUTRITION / RESILIENCE VALUE → NEED processing + procurement + demand

CORE CLAIM
- DRYLAND SUITABILITY
- LOWER WATER EXPOSURE

MECHANISM
- NUTRITION / RESILIENCE VALUE
- NEED processing + procurement + demand

CONSEQUENCE / LIMIT
- DRYLAND SUITABILITY
- LOWER WATER EXPOSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUTRITION / RESILIENCE VALUE; NEED processing + procurement + demand.

STAGE 08 — WATER-CROP LOCK-IN

WATER-CROP LOCK-IN · PROCUREMENT + CHEAP PUMPING · WATER-INTENSIVE CROP CHOICE · SOIL STRESS · DIVERSIFICATION BECOMES HARDER

CORE CLAIM
- PROCUREMENT + CHEAP PUMPING
- WATER-INTENSIVE CROP CHOICE

MECHANISM
- AQUIFER / SOIL STRESS
- DIVERSIFICATION BECOMES HARDER

CONSEQUENCE / LIMIT
- PROCUREMENT + CHEAP PUMPING
- WATER-INTENSIVE CROP CHOICE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER-INTENSIVE CROP CHOICE; AQUIFER / SOIL STRESS; DIVERSIFICATION BECOMES HARDER.

STAGE 09 — SCIENTIST-ENGINEER DISTINCTION

SCIENTIST-ENGINEER DISTINCTION · SLUICE GATES + BLOCK IRRIGATION · SEMI-DWARF CROP ADAPTATION · ASSURED WATER SYSTEMS · CROP SCIENCE · YIELD RESPONSE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
VISVESVARAYA → sluice gates + block irrigation	ENGINEERING → assured water systems	VISVESVARAYA → sluice gates + block irrigation
SWAMINATHAN → semi-dwarf crop adaptation	CROP SCIENCE → yield response	SWAMINATHAN → semi-dwarf crop adaptation

CORE CLAIM
- VISVESVARAYA → sluice gates + block irrigation
- SWAMINATHAN → semi-dwarf crop adaptation

MECHANISM
- ENGINEERING → assured water systems
- CROP SCIENCE → yield response

CONSEQUENCE / LIMIT
- VISVESVARAYA → sluice gates + block irrigation
- SWAMINATHAN → semi-dwarf crop adaptation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENGINEERING → assured water systems; CROP SCIENCE → yield response.

STAGE 10 — AGRARIAN TRADE-OFFS

AGRARIAN TRADE-OFFS · SECURE TENURE · OWNER CONCERNS · VULNERABLE-USER SAFEGUARDS · HIGH YIELD · FOOD SECURITY · ECOLOGICAL COST · MARKET RISK

SECURE TENURE → investment / owner concerns

CONSOLIDATION → scale / vulnerable-user safeguards → HIGH YIELD → food security / ecological cost → DIVERSIFICATION → resilience / market risk

CORE CLAIM
- SECURE TENURE → investment / owner concerns
- CONSOLIDATION → scale / vulnerable-user safeguards

MECHANISM
- HIGH YIELD → food security / ecological cost
- DIVERSIFICATION → resilience / market risk

CONSEQUENCE / LIMIT
- SECURE TENURE → investment / owner concerns
- CONSOLIDATION → scale / vulnerable-user safeguards

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH YIELD → food security / ecological cost.

STAGE 11 — AGRARIAN ANSWER SPINE

AGRARIAN ANSWER SPINE · DEFINE RIGHTS, RECORDS AND HOLDING STRUCTURE · TRACE TECHNOLOGY + WATER + MARKET PACKAGE · COMPARE REGIONAL OUTCOMES AND TRADE-OFFS · CONCLUDE WITH VIABLE DIVERSIFICATION · EVIDENCE LIMIT · USE STATE-SPECIFIC LAW AND PERIOD

DEFINE rights, records and holding structure → TRACE technology + water + market package → COMPARE regional outcomes and trade-offs → CONCLUDE with viable diversification → EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: Use state-specific law and period

CORE CLAIM
- DEFINE rights, records and holding structure
- TRACE technology + water + market package

MECHANISM
- COMPARE regional outcomes and trade-offs
- CONCLUDE with viable diversification

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: Use state-specific law and period,...
- DEFINE rights, records and holding structure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: Use state-specific law and period,.

EXTRA — SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · SECURE TENURE CAN IMPROVE INVESTMENT INCENTIVES, YET POORLY DESIGNED · LEASING REFORM CAN SEPARATE OWNERSHIP FROM EFFICIENT CULTIVATION WHILE · LAND RECORDS ARE FISCAL, CREDIT, PLANNING AND DISPUTE-RESOLUTION INFRASTRUCTURE · GREEN REVOLUTION GAINS CREATED FOOD SECURITY BUT ALSO PATH · CROP DIVERSIFICATION REQUIRES DEMAND, PROCESSING, INSURANCE AND LOGISTICS, NOT · ALLIED-SECTOR GROWTH CAN STABILISE INCOME BECAUSE PRODUCTION CYCLES AND · STATE REVENUE AND LAND DEPARTMENTS

OPTIONAL SOURCE DEPTH
- CAUTION: Secure tenure can improve investment incentives, yet poorly designed ceilings may
- CAUTION: Leasing reform can separate ownership from efficient cultivation while protecting
- CAUTION: Land records are fiscal, credit, planning and dispute-resolution infrastructure, not
- CAUTION: Green Revolution gains created food security but also path dependence in water,

ADVANCED DEBATE
- CAUTION: Crop diversification requires demand, processing, insurance and logistics, not price
- CAUTION: Allied-sector growth can stabilise income because production cycles and risks differ
- State revenue and land departments: implement land records, tenancy and ceiling
- Agricultural universities and extension systems: adapt seed and agronomy to local

LEGACY / NUANCE
- Procurement and irrigation agencies: shape crop profitability and regional
- Panchayats, co-operatives and farmer organisations: support local records, input
- CAUTION: Operation Barga is used to illustrate how political mobilisation and tenancy recording
- CAUTION: Punjab-Haryana cereal specialisation shows how irrigation and procurement can lock a

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels